OptiSAR-Net: A Cross-Domain Ship Detection Method for Multisource Remote Sensing Data

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***Abstract*— Optical and synthetic aperture radar (SAR) remote sensing are crucial for ship detection. Integrating SAR's all-weather imaging with optical data's shape recognition enhances downstream applications. However, current cross-domain methods often use unsupervised or semi-supervised techniques for single-source detection, limiting their practical use in cross-domain ship detection. Inspired by human visual cortex mechanisms, this article proposes OptiSAR-Net, an end-to-end cross-domain multisource ship detection network. Specifically, OptiSAR-Net features dual adaptive attention (DAA) for extracting standard features from SAR and optical images, and bilevel routing deformable spatial pyramid pooling-fast (BSPPF) for adapting to multiscale changes. To mitigate SAR noise, we employ VoV-GSCSP with spatial shuffling attention (VSSA) in the neck. OptiSAR-Net achieved state-of-the-art average precisions (APs) of 88.6% and 91.3% on the optical datasets DOTA and HRSC2016, respectively, and showed strong performance on the SAR datasets HRSID and SSDD. On the cross-domain heterogeneous dataset (CDHD), OptiSAR-Net differentiated ship targets effectively with only 2.7 million parameters and 11.7 GFLOPs, achieving an inference speed of 89 FPS on an NVIDIA RTX 3090. These results demonstrate that cross-domain multisource detection significantly enhances performance and application potential compared to single-source detection. Code is available at https://github.com/SCNU-RISLAB/OptiSAR-Net.**

***Index Terms*— Attention mechanism, cross-domain multisource detection, deep learning, lightweight, ship detection, synthetic aperture radar (SAR).**

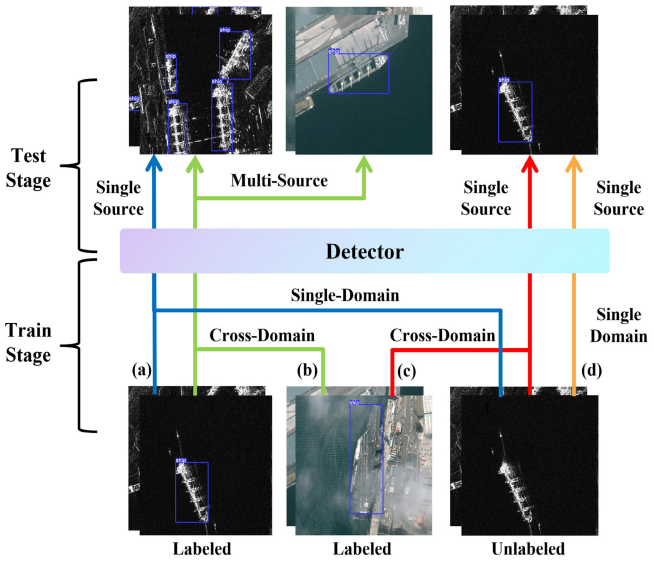


Fig. 1. Standard task modes for remote sensing ship detection. (a) Semisupervised single-domain SAR detection (blue path). (b) Cross-domain multisource detection (green path) (ours). (c) Unsupervised cross-domain SAR detection (red path). (d) Supervised single-source SAR detection (orange path). The supervised single-source optical detection is omitted in the image.

I. Introduction

OCEAN remote sensing technology advancements have provided abundant optical and synthetic aperture radar (SAR) images for ship detection [1]. Optical remote sensing images can intuitively capture information such as ship texture, color, and shape, but the data collection capability is limited by light and weather conditions. SAR sensors enable daily and all-weather detection, penetrate clouds and fog, and are unaffected by shadow obstruction and lighting conditions.

In recent years, ship detection technology driven by deep learning has significantly progressed. These studies have focused mainly on single-source image object detection, which has relatively poor model generalizability. Researchers are constantly exploring the integration of multisource image fusion with object detection to obtain target and scene information more effectively [2]. Existing cross-domain detection research on optical and SAR remote sensing images has focused mainly on unsupervised [3] or semisupervised [4] feature alignment, addressing the labor-intensive labeling challenge of SAR data through knowledge transfer to enable SAR single-source detection. However, in practical applications, multisource detection is critical in enhancing situational awareness capabilities. Fig. 1 compares different remote sensing ship detection tasks.

With the advancement of deep learning, future remote sensing techniques for ship inspection may not require distinguishing between optical and SAR data or considering varying imaging modes and resolutions; an end-to-end model will be sufficient [5]. Recent studies have shown that even

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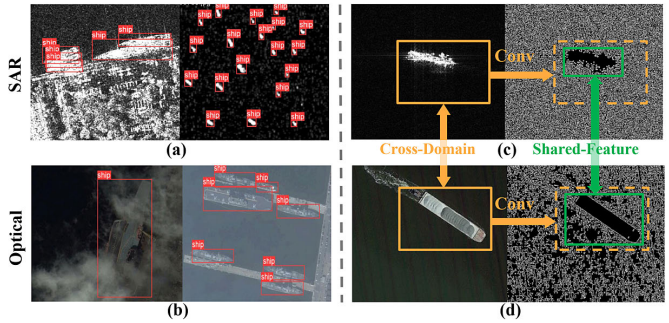


Fig. 2. (a) and (b) Present ship targets in SAR and optical remote sensing imagery, with ground truths delineated by red bounding boxes. The processed outcomes of dual-domain images (c) and (d), following grayscale transformation, contrast enhancement, and edge detection, reveal intrinsic commonalities across both images regarding the geometric shapes of the ships, variations in background contrast, and edge feature details.

mismatched image pairs can enhance ship detection and classification performance through multimodal fusion networks and transfer learning [6]. Cross-domain detection of multisource data can achieve complementary effects between optical and SAR data, enhancing ship detection capabilities [7]. However, detecting cross-domain multisource ships poses a new, highly challenging problem. The imaging mechanisms of the two frequency domain data types differ significantly, as shown in Fig. 2(a) and (b). SAR images are influenced by the roughness of the Earth's surface and radiation conditions, resulting in minimal variations in the brightness of ship targets with a lack of distinct textures. In contrast, optical images are more complex due to varying lighting conditions, presenting ships in different colors and richer textures. Moreover, both frequency domains encounter significant challenges in detecting the multiscale variations inherent to ship targets. It is difficult for a model trained with the same weights to extract compelling features evenly from images of different classes, which leads to a lack of generalizability of many detection algorithms across different datasets.

Nevertheless, the geometric shape and contrast of objects are identified as the most critical recognition features, independent of the frequency domain background. As illustrated in Fig. 2(c) and (d), optical images utilize reflected visible light, creating a distinct visual contrast with the smooth water texture in the background. Although the texture and contrast in SAR images differ from those in optical images, ships are prominently displayed in radar echoes due to their high reflectivity from metallic structures and multisurface reflections. This commonality indicates the potential for multisource recognition in multiple domains.

Therefore, we propose a lightweight cross-domain detection model for ship detection in multisource optical and SAR images. First, to address the perceptual differences caused by the different imaging mechanisms of images, we introduce the dual adaptive attention module (DAAM). Additionally, we propose bilevel routing deformable spatial pyramid pooling-fast (BSPPF) to address scale variations in the two types of images. Finally, the model mitigates noise effects in SAR images using VoV-GSCSP with spatial shuffling attention (VSSA) in the neck. Our work aims to simulate the guidance of the human brain's prefrontal cortex, directing the model to focus on ship-related visual features rather than the varying frequency domain backgrounds to generalize consistent shape, texture, or contrast information. To the best of our knowledge, our proposed OptiSAR-Net is among the first to discuss cross-domain multisource remote sensing ship detection methods. In summary, the main contributions of this article are as follows.

1) OptiSAR-Net for multisource detection task. We propose OptiSAR-Net, cross-domain ship detector designed for detecting ships in both SAR and optical remote sensing images. Extensive experiments were conducted on both single-domain and cross-domain datasets. OptiSAR-Net achieved SOTA results on the DOTA (88.6%) and HRSC2016 (91.3%) datasets and demonstrated strong competitiveness on the HRSID (90.2%) and SSDD (97.6%) datasets. The model effectively facilitates multisource ship detection on the constructed cross-domain heterogeneous dataset (CDHD).
2) Innovative attention mechanisms. OptiSAR-Net incorporates three novel plug-and-play attention mechanisms: Dual adaptive attention (DAA), bilevel routing deformable attention (BRDA), and spatial shuffle attention (SSA). These three modules collectively enhance the interpretability of contextual understanding and cross-domain feature fusion, leveraging the complementary advantages of optical and SAR images to help the model achieve reliable ship imaging under all weather conditions.
3) Lightweight and efficient design. OptiSAR-Net is an end-to-end lightweight model that meets the resource constraints of practical satellite terminals. With a compact size of only 2.7 million parameters and 11.7 GFLOPs, reaching 89.8 frames per second (FPS) on an NVIDIA RTX 3090.

II. Related Work

A. Single-Source Remote Sensing Ship Detection

In the early days, the constant false alarm rate (CFAR) achieved SAR ship detection by establishing a clutter distribution model based on local ocean clutter data. Traditional optical image methods often emphasize feature design. Leng et al. [8] fully considered the imaging characteristics of SAR images, distinguishing between ships and radio frequency interference (RFI) through noncircularity and non-Gaussianity. This represents an exploration in the nondeep learning domain.

With advanced deep learning, convolutional neural networks (CNNs) have demonstrated their powerful representation capabilities in ship detection. Zhang et al. [9] introduced scattering features into the deep learning framework and proposed the multiscale global scattering feature association network (MGSFA-Net) for ship target recognition in SAR images. Cheng et al. [10] optimized a new objective function by imposing regularization constraints during training to effectively address object rotation changes in optical images. The more extensive parameters of similar methods pose challenges for end deployment. Improved SA-YOLOv5 [11] addresses intraclass imbalances in SAR ship detection for better

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interpretability. AF-OSD [12] offered solutions for challenges in optical ship rotation detection, progressively increasing speed and accuracy. These advancements have improved ship detection in terms of timeliness, accuracy, and robustness; however, limitations persist owing to the inherent characteristics of single-source sensor data.

B. Cross-Domain Ship Detection

By fully considering multisource data information from different sensors, it is possible to achieve ship detection under varying weather and lighting conditions in practical applications. Current cross-domain detection methods for distinguishing between optical and SAR images focus primarily on addressing the scarcity of SAR labels by learning SAR ship features from annotated optical images. Yuan et al. [3] introduced OmniAdapt, a novel unsupervised domain adaptation (UDA) architecture, to address the issue of insufficient labels for cross-domain SAR ship detection. Zheng et al. [13] suggested that a few labeled SAR samples are available and proposed a dual-teacher framework to address the mutual interference between optical supervision and SAR supervision, exploring semisupervised cross-domain ship detection (SCSD).

To our knowledge, we are among the first to conduct multisource ship detection in the cross domain of optical and SAR.

C. Attention Mechanism

The attention mechanism is a technique inspired by the human visual attention system, enabling models to focus on crucial information in input sequences and enhancing overall performance. Sun et al. [14] proposed a dual-branch feature fusion (D-BFF) method based on improved transformer blocks to achieve high-precision ship detection under recomplex SAR images. For cross-domain image detection, Li et al. [15] introduced an innovative object detection architecture; it extracts detailed and rich features from optical images. Then, it recalibrates the spatial attention weights assigned to optical and SAR features based on these feature maps. These studies have demonstrated that cross-domain detection models outperform single-domain detection.

III. Method

This section provides a detailed overview of the proposed method. Compared to YOLOv10 [16], we have chosen the more stable YOLOv8-n as the baseline model for OptiSAR-Net. The overall framework of OptiSAR-Net is illustrated in Fig. 3. Initially, we replace the original C2f in the backbone with the DAAM, enabling the model to extract cross-domain features from multiple sources dynamically. Subsequently, we introduce the BRDA, which uses BSPPF to further increase the model's robustness in handling scale variations of ship targets. Finally, we incorporate the VSSA in the neck to reduce the model parameters and complexity while filtering out noise interference from SAR images. These three attention design principles proposed in this paper apply equally to other object detection tasks, particularly in cross-domain multisource detection scenarios.

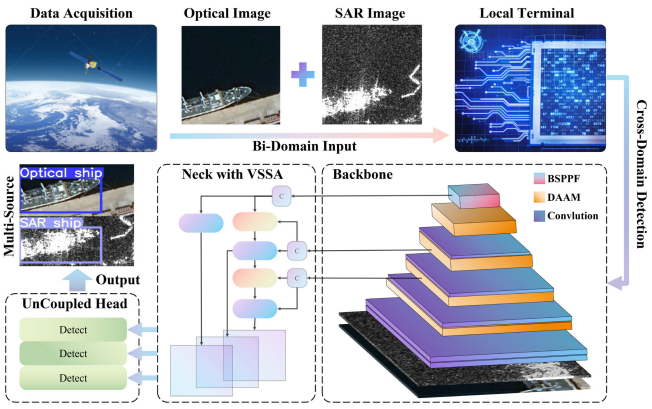


Fig. 3. General framework diagram of OptiSAR-Net. In practical applications, satellite sensors can collect multisource (or single-source) data, which can be directly input into models for on-site inference. This ultimately provides multidomain (or single-domain) detection results, enabling all-weather ship detection. Note that OptiSAR-Net's single-domain detection performance surpasses that of other models trained on single-source data (benefiting from the complementary effect of multisource training).

A. Dual Adaptive Attention Module

Similar to how the human visual system converts all external signals into electrical signals and processes them through the visual cortex, images from different frequency domains are unified as digital signals (matrices) in deep learning and are essentially the same. Additionally, ships in both SAR and optical images share common characteristics regarding geometric shape, background contrast, and edge information. Therefore, as long as the model, like the human brain's prefrontal cortex, can maintain sufficient focus on ship-related visual features, it can effectively distinguish ships across different domains. Due to the complementary advantages, the model's discriminative capability will surpass that of single-source training.

Introducing attention mechanisms into the model is crucial for achieving multisource image fusion feature detection. Kernel selection is an adaptive and effective dynamic context modeling technique. SKNet [17] introduces multiple branches with different convolutional kernels and selectively combines them along the channel dimension. ResNeSt [18] extends SKNet by partitioning input feature maps. LSK [19] relies on large kernels to adaptively aggregate ample kernel information in spatial dimensions, mimicking the spatial variances of different targets in the image space. In contrast, the DAA performs secondary modeling of long-distance context through dilated convolutions with residual connections, effectively integrating standard features of cross-domain images. The corresponding structural comparison results can be found in Fig. 4.

To adaptively model longer distance cross-domain image features, we decompose large kernel convolutions into two identical small kernels with different dilation rates to achieve context awareness and fit features of different receptive fields. A vast receptive field can lead to the loss of small target features; hence, we control the total receptive field at the theoretical value of 9. The output features with a large receptive field X_2 are further utilized in the subsequent

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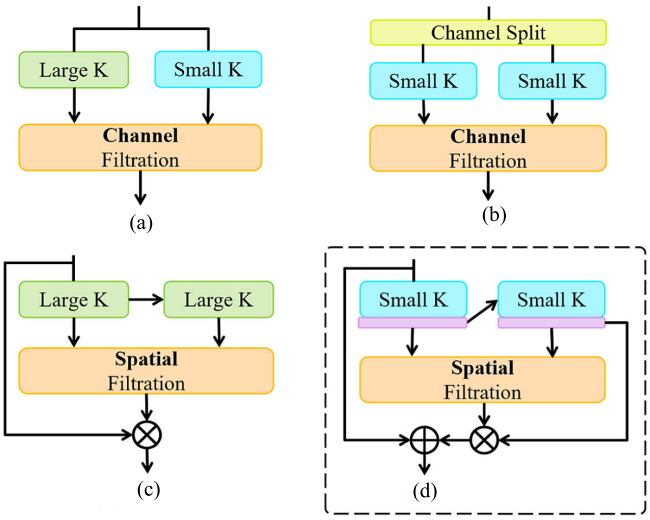


Fig. 4. Structural comparison between the DAA and other representative selection mechanisms. The purple bar represents the BN+SiLU operation. K represents the kernel. (a) SKNet. (b) RESNeSt. (c) LSK. (d) DAA (ours).

attention calculations

$$k_1 = k_2 = 3; d_1 = 1, d_2 = 3 \tag{1}$$

$$\mathrm{RF}_1 = k_1, \mathrm{RF}_2 = d_2(k_2 - 1) + \mathrm{RF}_1 = \mathrm{RF}\left(\mathcal{F}^{9\times 9}\right) \tag{2}$$

$$X_2 = \widetilde{\mathcal{F}}_2\left(\widetilde{\mathcal{F}}_1(\mathrm{X})\right). \tag{3}$$

The kernel size k_i, dilation rate d_i, and receptive field RF_i are defined for the ith standard convolution. $\widetilde{\mathcal{F}}_i(\cdot)$ adds batch normalization (BN) and SiLU operations after $\mathcal{F}(\cdot)$ to prevent model degradation. Decomposing large kernel convolutions for secondary attention calculation provides explicit features and resolves the computational burden of large kernel convolutions

$$\widetilde{\mathrm{V}}_i = \widetilde{\mathcal{F}}_{\mathrm{i}}^{1\times 1}(\mathrm{X}_i), i = 1, 2. \tag{4}$$

Aggregating attention queries with different receptive fields and capturing heterogeneous standard features can be achieved through functions $\phi_{\mathrm{mean}}(\cdot)$ and $\phi_{\mathrm{max}}(\cdot)$. After activation, preliminary spatial embedding weights are obtained, allowing for the adaptive selection of spatial features via attention; each is subsequently weighted with the corresponding spatial selection masks for the secondary attention queries, and after convolutional fusion, the final attention weights $\widetilde{Q}$ are derived

$$\widetilde{\mathrm{A}} = \left[\widetilde{\mathrm{A}}_1; \widetilde{\mathrm{A}}_2\right] \tag{5}$$

$$\mathrm{Q} = \sigma\left(\widetilde{\mathcal{F}}^{1\times 1}\left(\phi_{\mathrm{mean}}\left(\widetilde{\mathrm{A}}\right); \phi_{\mathrm{max}}\left(\widetilde{\mathrm{A}}\right)\right)\right) \tag{6}$$

$$\widetilde{\mathrm{Q}} = \widetilde{\mathrm{A}}_1 \cdot \mathrm{Q}[:0::] + \widetilde{\mathrm{A}}_2 \cdot \mathrm{Q}[:1::] \tag{7}$$

where $\sigma(\cdot)$ represents the sigmoid function, and $Q[:i::]$ corresponds to the ith spatial embedding weight. Finally, the output is obtained via elementwise multiplication of the attention mechanism with the retained global features.

Fig. 5 illustrates the structure of the DAAM. We visually demonstrate how the DAA adaptively extracts and fuses cross-domain image features through secondary attention while also describing how the DAAM addresses feature disappearance in SAR images through residual connections.

B. Bilevel Routing Deformable Spatial Pyramid Pooling-Fast

Spatial pyramid pooling-fast (SPPF) [20] extracts image features at different scales. However, its formulation, which captures global contextual information in images, overlooks the correlations between scales. Inspired by bilevel routing attention (BRA) [21] and deformable attention [22], we propose a new dynamic sparse and offsettable attention mechanism called BRDA to construct BSPPF. The key idea is to apply attention mechanisms to spatial features before multilevel pooling, modeling global contextual information in an offsettable manner. Attention computation is performed only on focused candidate regions, allowing for flexible capture of crucial region information. This approach effectively mitigates the computational overhead of global attention calculations while mitigating the background feature differences caused by different frequency domains. We detail the workflow of the BRDA in Fig. 6. To simplify notation, we discuss the case of single-head self-attention only.

1) Query and Offset Reference Points Generation: Given a 2-D input feature map $\mathrm{X} \in \mathbb{R}^{\mathrm{H}\times\mathrm{W}\times\mathrm{C}}$, we only use it to generate query Q

$$\mathrm{W}^{\mathrm{q}} = \mathcal{F}^{1\times 1}, \mathrm{Q} = \mathrm{X}\mathrm{W}^{\mathrm{q}}. \tag{8}$$

A uniform grid of reference points denoted by $r \in \mathbb{R}^{\mathrm{W}\times\mathrm{H}\times 2}$ was subsequently generated. To effectively relate these reference points to critical regions in the image, we perform convolutional operations $\tilde{O}(\cdot)$ on the query subnetwork to generate offset adjustments. Specifically, the query first underwent a 7 × 7 large kernel convolution to capture locally related features. Notably, we employed BN to consider features across all channels deeply, enabling the learning of uniform offset adjustments. Finally, employing GELU activation and a 1 × 1 convolution yielded 2-D offsets $\Delta\mathrm{r}$. To ensure the interpretability of the offset features, we utilized the bilinear interpolation method $\Gamma(\cdot)$ to sample the input feature map

$$\Delta\mathrm{r} = \tilde{\mathrm{O}}(\mathrm{Q}), \tilde{x} = \Gamma(x; \theta(\mathrm{r} + \Delta\mathrm{r})) \tag{9}$$

where $\theta(\cdot)$ denotes the clamping operation, which restricts the offsets within $[-0.5, 0.5]$ to prevent excessive feature shifting.

2) Candidate Region Generation: Unlike conventional attention mechanisms, we generate key-value pairs from resampled feature maps and jointly divide them with queries into $S \times S$ nonoverlapping regions as candidate regions, each containing (HW/S^2) feature vectors

$$\widetilde{\mathbf{K}} = \tilde{x}W_k, \widetilde{\mathbf{V}} = \tilde{x}W_{\mathrm{v}} \tag{10}$$

where $\mathbf{Q}$, $\widetilde{\mathbf{K}}$, and $\widetilde{\mathbf{V}} \in \mathbb{R}^{s^2\times(HW/s^2)\times c}$. By utilizing routing computation TopkRouting $(\cdot)$, we deduce the degree of correlation between regions to obtain an attention index matrix. For each candidate region, only the four most relevant connections are retained for fine-grained tokenization, and key-value pairs K and V are selectively gathered via the method KVGather $(\cdot)$

$$\mathrm{I} = \mathrm{TopkRouting}\left(\mathrm{Q}\left(\widetilde{\mathrm{K}}\right)^T\right) \tag{11}$$

$$\widetilde{\mathrm{K}}^g, \widetilde{\mathrm{V}}^g = \mathrm{KVGather}\ (\mathrm{K}, \mathrm{V}, \mathrm{I}). \tag{12}$$

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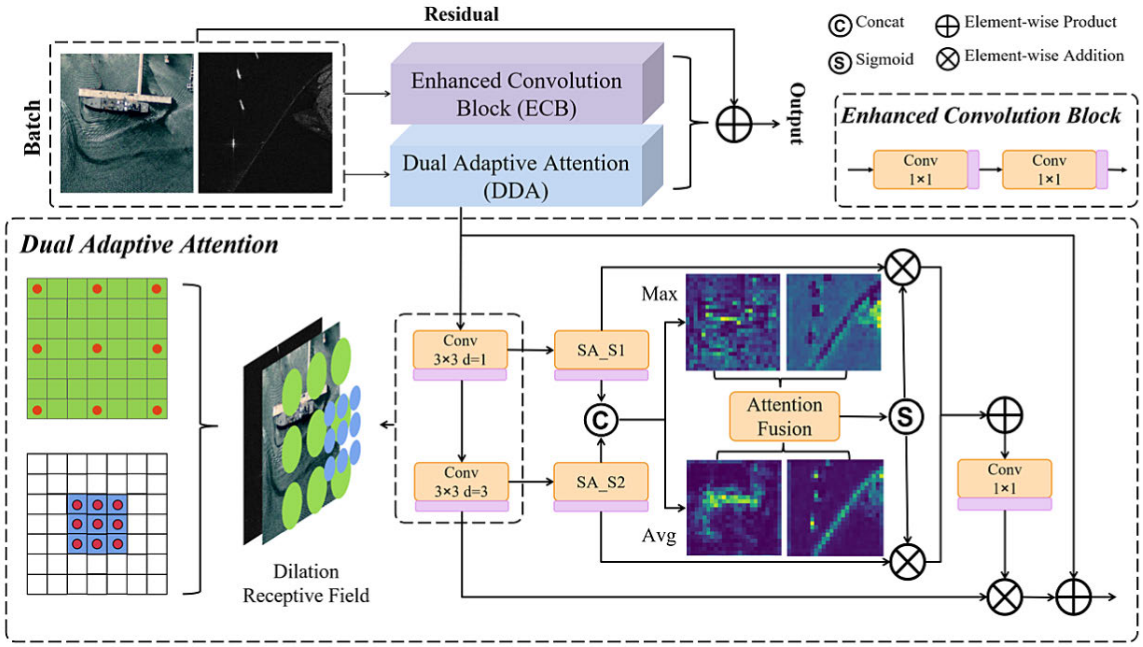


Fig. 5. Structure of DAAM. The DAA is responsible for dynamically extracting standard features from input batch images. Additionally, we emphasize the importance of residual connections. By introducing the original input and separately enhancing features through convolutional blocks in the output, we aim to retain as much critical data from the original images as possible. This operation effectively balances the information richness of optical and SAR images, preventing overfitting in SAR images caused by deep convolution.

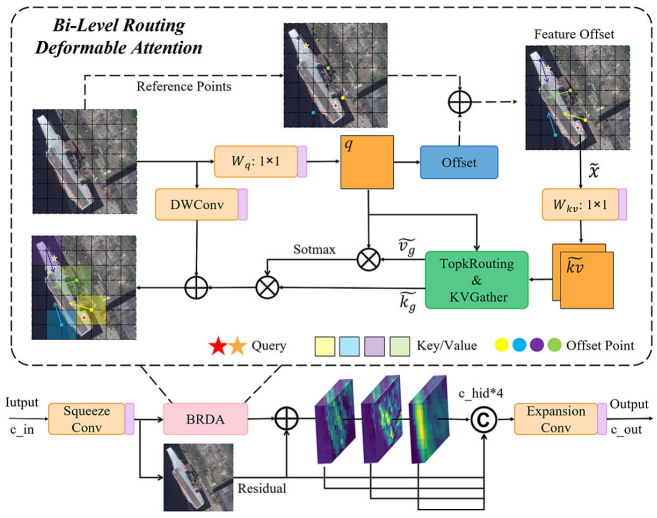


Fig. 6. Structure of BSPPF. During the backpropagation process, BRDA can provide offset compensation for targets of different scales, thereby obtaining more effective key-value pairs.

3) Attention Calculation With Residual Connections: Self-attention calculations are performed on the collected key-value pairs in a conventional manner. We have additionally introduced a 3 × 3 depthwise convolution (DWConv) [23] to establish residual connections on the resampled values, ensuring reliable gradients for BRDA and accelerating model convergence

$$\mathrm{Z} = \mathrm{Attention}\left(\mathbf{Q}, \widetilde{\mathbf{K}}^{g}, \widetilde{\mathbf{V}}^{g}\right) + \mathcal{F}_{\mathrm{DW}}^{3\times 3}\left(\widetilde{\mathbf{V}}\right). \tag{13}$$

The BSPPF employs dynamic deformable attentive queries to optimize spatial allocation in pyramid pooling. It accurately captures contextual information surrounding targets, providing more reliable multiscale features for enhanced feature enrichment. BRDA uses sparsity to save computation and memory, involving only GPU-friendly matrix multiplication.

C. VoV-GSCSP With SSA

Real-time detection at satellite terminals places a high demand on lightweight model design. GSConv [24] uses shuffling to infuse features generated by standard convolutions into each part of the features generated by depthwise separable convolutions (DSCs) [23], effectively reducing computational costs while preserving connections between channels as much as possible. Although the performance of this approach closely matches that of standard convolutions, it struggles to handle noise interference in SAR images. Unlike optical images with rich pixel information, noise from high-dimensional features in the neck region of SAR images significantly disrupts feature fusion in models. Therefore, we propose spatial shuffling attention (SSA), a mechanism that efficiently captures pixel-level pairwise relationships, suppressing the model's focus on noisy parts of the feature map with minimal introduction of additional parameters. The constructed VSSA is applied to the neck region.

As shown in Fig. 7, the given feature map $\mathrm{X} \in \mathbb{R}^{\mathrm{C}\times\mathrm{H}\times\mathrm{W}}$ is divided into g groups along the channel dimension and shuffled separately. By projecting the input vectors into multiple subspaces and performing self-attention operations in each subspace, the model can learn features and dependencies in different spaces in parallel, enhancing its ability to discriminate against noise

$$X = \left[X_1, \cdots, X_g\right], X_k \in \mathbb{R}^{\mathrm{C/g}\times\mathrm{H}\times\mathrm{W}} \tag{14}$$

$$C = \left[c^{(1)}, c^{(2\to\langle \mathrm{N}-1\rangle)}, \ldots, c^{(\langle \mathrm{N}-1\rangle\to 2)}, c^{(\mathrm{N})}\right]. \tag{15}$$

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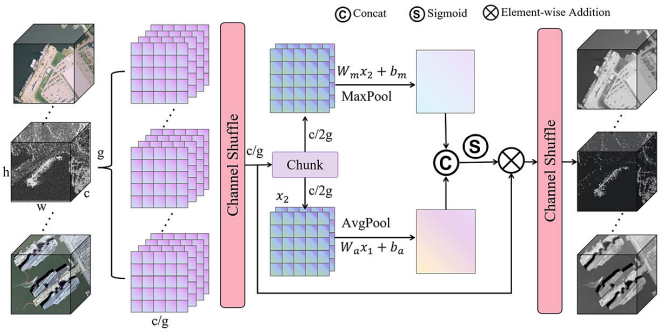


Fig. 7. Structure of the VSSA. For a grouped subnetwork, the SSA calculation involves the four parameters $-W_a$, b_a, W_m, and b_m-, indicating its lightweight attention mechanism.

The SSA divides the tensor into smaller subspaces for each subfeature to perform spatial attention calculations separately. Before this, global average pooling (GAP) embeds global information in one channel. In contrast, global maximum pooling (GMP) embeds critical features in the other channel. The corresponding parts are enhanced via a univariate linear regression function and concatenated to form the final attention representation. Dual-path spatial attention focuses on which part contains noise information and which part contains valuable information

$$\mathrm{X_{i1}}, \mathrm{X_{i2}} = \mathrm{Chunk}(\mathrm{X_i}), i = 1, \ldots, \mathrm{g} \tag{16}$$

$$X'_{i1} = W_a \cdot \mathrm{P_{gap}}(\mathrm{X_{i1}}) + b_a, X'_{i2} = W_m \cdot \mathrm{P_{gmp}}(\mathrm{X_{i2}}) + b_m \tag{17}$$

$$X'_i = \left[X'_{i1}; X'_{i2}\right] \tag{18}$$

where $\mathrm{X_{i1}}, \mathrm{X_{i2}} \in \mathbb{R}^{\mathrm{C}/2\ \mathrm{g}\times\mathrm{H}\times\mathrm{W}}$; $W_a, \mathrm{W_m} \in \mathbb{R}^{C/2G\times1\times1}$; and $b_a, b_m \in \mathbb{R}^{C/2G\times1\times1}$. After the filtered feature map is obtained, the channels are shuffled once again to restore their original semantic representations. At this point, the spatial features of each channel are more expressive.

IV. Experiment

In this section, ablation studies were conducted on the three attention mechanisms proposed in this article, using the baseline model as a foundation. Additionally, these investigations were supplemented with further ablation experiments on Faster-RCNN [25] to validate the transferability of the methodologies. For comparative analysis, the proposed approach was juxtaposed with classical SOTA models and representative frameworks from related domains across five datasets (four single-domain datasets and one cross-domain dataset). This comprehensive evaluation demonstrates the proposed method's exceptional versatility and cross-domain multisource detection capabilities.

A. Experimental Preparation

1) Dataset: DOTA [26] is a large-scale aerial image dataset of 2806 images annotated by experts with 15 object categories. Using official development tools, we cropped the images to 1024×1024 with an overlap of 100 and a threshold of 0.7 for incomplete objects. We selected 800 cropped images containing ships (a total of 20753) for the horizontal bounding box (HBB) task. HRSC2016 [27] is a challenging dataset for detecting ships with high aspect ratios and arbitrary orientations in aerial images. It was suitable for verifying the validity of the model on optical images. The HRSID [28] dataset consists of 5604 images. We segmented 5604 cropped SAR images at a ratio of 6.5:3.5 for the training and test sets. The SSDD dataset [29] contains imaging results from different satellite sensors. We constructed a CDHD to validate the multisource detection effectiveness of the model. This dataset retains all original images and labels from HRSC2016 and includes 416, 181, and 444 images extracted from HRSID, which are added to the training, validation, and test sets, respectively. The ratio of optical images to SAR images is 1:1. Table I shows the relevant datasets in detail.

TABLE I
Basic Information and Experimental Settings of the Four Experimental Datasets

	DOTA	HRSC2016	HRSID	SSDD	CDHD
Images size	1024×1024	800×800	800×800	640×640	800
Train:Valid:Test	2:1:1	4.1:1.7:4.2	6.5:3.5:0	8:2:0	4.1:1.7:4.2
Number of class	1	1	1	1	1
Batch size	16	32	32	32	32
Epoch	100	100	100	100	100
Image type	Optical	Optical	SAR	SAR	Optical&SAR
Number of image	800	1061	5604	1160	1061+1061

2) Experimental parameters: We used an AMD EPYC 7601 32-Core Processor, Ubuntu 20.04, and an NVIDIA RTX 3090 graphics card to build the proposed modeling method, which is based on CUDA version 12.2 and the PyTorch framework. In the experiment, we standardized the training parameters of different models. The momentum was set to 0.937, the momentum decay coefficient was set to 0.0005, and the initial learning rate was set to 0.01.
3) Assessment criteria: FLOPs and parameters were chosen as measures of the model size. Additionally, we highlighted our lightweight design by adding FPS as a metric for evaluating the inference time. FPS represents the number of image frames the model can process per second during the inference phase. In this article, AP_{50} is the AP computed at an IoU threshold of 0.5 for single-object detection. Moreover, $AP_{50:95}$ (mAP) denotes the average AP computed at different IoU thresholds ranging from 0.5 to 0.95 (with a step size of 0.05).

B. Ablation Experiment

The ablation experiments of OptiSAR-Net were first conducted. DAAM, BSPPF, and VSSA were sequentially replaced in the baseline model. The experimental results are presented in Table II, where the model was trained and validated on the CDHD dataset. Combining any two or all three enhancements outperformed using them individually, demonstrating the synergistic effects of attention mechanisms. When all enhancements were incorporated, the model improved ship

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TABLE II
ABLATION STUDIES OF THREE IMPLEMENTS ON THE BASELINE MODEL

Method			OPTICAL (HRSC2016)		SAR (*HRSID)		ALL		#Param. (M)	FLOPs (G)
DAAM	BSPPF	VSSA	AP_{50}^{val}	$AP_{50:95}^{val}$	AP_{50}^{val}	$AP_{50:95}^{val}$	AP_{50}^{val}	$AP_{50:95}^{val}$		
	YOLOv8-n		90.8	74.2	78.7	51.5	84.8	62.9	3.0	12.6
✓			90.9	75.4	80.8	53.9	85.9	64.7	2.9	12.5
	✓		91.2	76.0	80.5	53.9	85.9	65.0	3.1	12.7
		✓	91.3	75.9	80.3	53.1	85.8	64.5	2.8	11.9
✓	✓		91.9	76.5	81.4	54.1	86.7	65.3	3.0	12.6
	✓	✓	91.6	76.3	81.5	53.8	86.6	65.0	2.9	11.9
✓		✓	91.4	76.6	79.6	52.2	85.5	64.4	**2.6**	**11.7**
✓	✓	✓	**91.8**	**76.7**	**82.1**	**54.7**	**87.0**	**65.7**	2.7	11.8

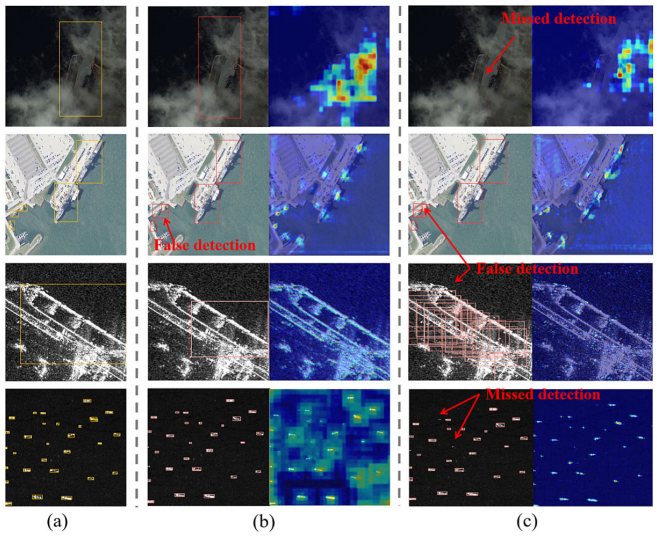


Fig. 8. Comparison of gradient-weighted class activation mapping visualization results on the CDHD dataset. (a) Ground truth. (b) OptiSAR-Net. (c) YOLOv8-n. Our approach demonstrates significantly superior performance in cross-domain multisource detection.

TABLE III
ABLATION EXPERIMENTS FOR BSPPF

Model	OPTICAL (HRSC2016)		SAR (*HRSID)		ALL		#Param. (M)	FLOPs (G)
	AP_{50}^{val}	$AP_{50:95}^{val}$	AP_{50}^{val}	$AP_{50:95}^{val}$	AP_{50}^{val}	$AP_{50:95}^{val}$		
SPPF (base)	91.4	76.6	79.6	52.2	85.5	64.4	**2.6**	**11.7**
+BRA	91.6	76.7	80.5	52.9	86.1	64.8	**2.6**	**11.7**
+MaxPool	91.5	76.6	80.8	53.1	86.2	64.9	**2.6**	**11.7**
+QKVConv	91.6	76.7	81.7	53.4	86.7	65.1	2.7	11.8
+Offset	**91.8**	**76.8**	81.9	54.5	86.9	**65.7**	2.7	11.8
+LWin (ours)	**91.8**	76.7	**82.1**	**54.7**	**87.0**	**65.7**	2.7	11.8

TABLE IV
ABLATION STUDIES OF THREE IMPLEMENTS ON FASTER-RCNN

Method			OPTICAL (HRSC2016)		SAR (*HRSID)		ALL		#Param. (M)	FLOPs (G)
DAA	BRDA	SSA	AP_{50}^{val}	$AP_{50:95}^{val}$	AP_{50}^{val}	$AP_{50:95}^{val}$	AP_{50}^{val}	$AP_{50:95}^{val}$		
Faster R-CNN			90.9	69.3	67.4	46.6	78.9	58.0	**41.3**	**134.0**
✓			91.7	70.5	68.2	47.0	80.0	58.8	42.8	137.6
✓	✓		92.3	72.8	70.4	49.2	81.4	61.0	43.1	138.5
✓	✓	✓	**93.5**	**72.9**	**71.8**	**50.4**	**82.7**	**61.7**	43.1	138.9

TABLE V
COMPARISON WITH OTHER MODELS ON DOTA

Model	AP_{50}^{val} (%)	$AP_{50:95}^{val}$ (%)	#Param. (M)	FLOPs (G)	$FPS_{b=1}^{3090}$ (f/s)
YOLOv8-n (base)	87.5	56.4	3.0	21.0	**104.0**
YOLOv5-n	86.2	53.6	**2.5**	**18.4**	100.6
YOLOv9-t [30]	80.5	50.3	2.6	27.4	82.2
YOLOv10-n [16]	85.7	54.2	2.6	19.9	88.7
YOLOX-t [31]	72.8	50.2	5.0	19.4	75.2
RTMDet-t [32]	79.3	54.8	4.9	20.5	79.3
PPYOLOE-Plus-s [33]	85.4	56.8	7.9	20.3	68.9
Faster R-CNN [25]	68.6	47.9	41.3	209	55.8
OptiSAR-Net (ours)	**88.6**	**58.5**	2.7	19.5	89.5

classification performance in optical and SAR images, particularly in distinguishing ships in SAR images. Additionally, the model benefited from lightweight improvements. The results show that the designed ship detection method in cross-domain multisource scenarios is reliable. The model exhibited sufficient attention to discern and enhance target recognition across different frequency domains rather than failing to converge.

Fig. 8 visualizes the model's attention regions. For YOLOv8-n, which lacks dynamic attention properties, frequent feature variations in the dataset prevented it from learning appropriate weights during training. This limitation hindered its performance in effectively handling cross-domain image detection and varying ship scales. In contrast, the dynamic nature of OptiSAR-Net adeptly addressed this issue.

BRDA flexibly differentiated the importance of features at different scales at the junction between the backbone and the neck, enabling adaptive selection among multiple features. Table III provides a detailed account of the BRDA design process. Here, "base" referred to "YOLOV8-n+DAAM+VSSA." Furthermore, the other five experimental sets involved incorporating BRA, downsampling via maxpool when generating key-value pairs, sampling subspaces through convolutional forms, allowing attention to feature displacements, and using large windows to generate candidate regions. Notably, when resampling features through reference point compensation, the selected candidate regions were closer to the actual targets, leading to a significant increase in accuracy.

Furthermore, as depicted in Table IV, we incorporated the proposed three attention into the backbone, backbone tail, and RPN of Faster-RCNN [25]. The introduction of our designed attention mechanisms significantly enhances the model's sensitivity to cross-domain images.

C. Comparison Experiment

Beyond conducting multisource detection performance comparison experiments on cross-domain datasets, we expanded the analysis to include four single-domain datasets. This expansion aimed to showcase the exceptional generalization capabilities of OptiSAR-Net. The selected datasets comprised two prevalent optical datasets (DOTA and HRSC2016) and two SAR datasets (HRSID and SSDD). For comparison, we chose models renowned in remote sensing or those representing SOTA detectors introduced in recent years.

Table V presents the comparative results on the DOTA dataset. OptiSAR-Net achieved SOTA performance with a mAP of 0.585, surpassing YOLOv8-n by 3.6%. When confronted with dense and small-scale ship targets in DOTA,

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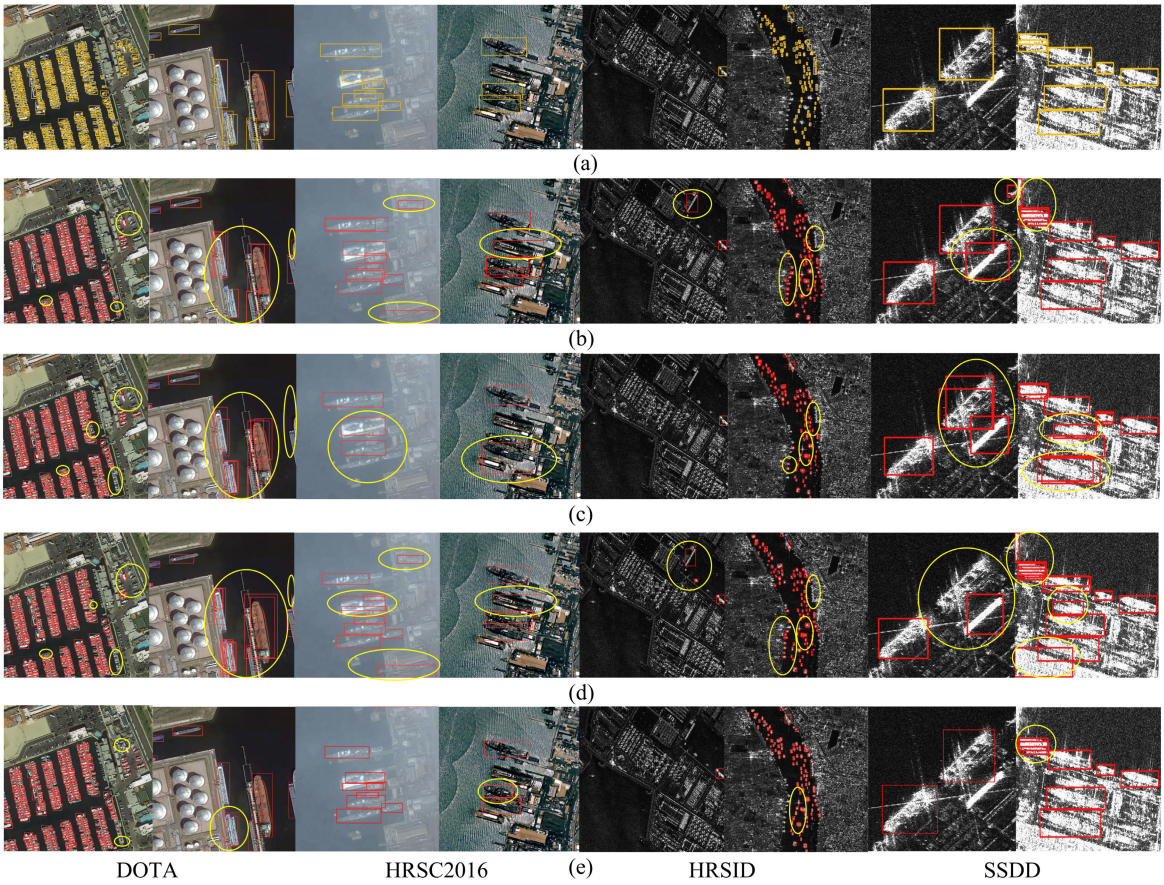


Fig. 9. Visual evaluation results of our method were compared with those of representative models on the DOTA, HRSC2016, HRSID, and SSDD datasets. (a) ground truth, (b) YOLOv8-n, (c) YOLOv10-n, (d) Faster-RCNN, and (e) OptiSAR-Net. The yellow circles represent missed or misclassified ship targets. For single-source detection task, dynamic attention can adaptively capture relationships between image features, concentrating computations on key regions.

BSPPF shifts the model's attention to lower level features, thereby capturing more small-scale targets. The visual evaluation results, as shown in Fig. 9, demonstrate that OptiSAR-Net is the model with the lowest error rate. Although the latest detectors represented by YOLOv10 [16] have shown good performance on large public datasets such as COCO, they have not achieved the desired results in the context of remote sensing. Owing to the unavoidable matrix transformation computations brought by attention mechanisms, OptiSAR-Net does not excel in inference speed but remains within an acceptable range. YOLOv5-n has the fewest parameters among these models, while YOLOv8-n achieved the fastest inference speed.

Table VI shows the comparison results of our model with other models on the HRSC2016 dataset. We selected relevant methods in the remote sensing domain for comparison, such as R3Det [34]. OptiSAR-Net once again achieved SOTA performance. In dealing with challenges such as haze occlusion and complex sea clutter interference, SSA effectively filters out feature noise, enabling more precise ship detection. While two-stage models such as Faster R-CNN [25] have shown decent results, it is evident that they need help in overcoming the resource constraints of practical endpoints.

Table VII presents the experimental results on the HRSID dataset. The asterisk (*) denotes the evaluation values. Owing to the limitations of attention mechanisms in SAR images lacking semantic information, our approach did not outperform the latest SAR detector, SA-YOLOv5-Plus [11], in the single-source SAR task. SA-YOLOv5-Plus [11] achieved an impressive accuracy of 94.8%, but its model size is not comparable to that of OptiSAR-Net. Compared with the similarly sized DBW-YOLO [38], OptiSAR-Net still demonstrates strong competitiveness.

Table VIII displays the results of the comparative experiments on the SSDD dataset. SSDD, being a small-scale dataset, presents lower difficulty than larger datasets do, leading to decent performance by most models. Attention

TABLE VI
COMPARISON WITH OTHER MODELS ON HRSC2016

Model	AP_{50}^{val} (%)	$AP_{50:95}^{val}$ (%)	#Param. (M)	FLOPs (G)	$FPS_{b=1}^{3090}$ (f/s)
YOLOv8-n (base)	89.9	72.9	3.0	12.8	**110.3**
YOLOv5-n	89.1	69.3	**2.5**	**11.2**	106.2
YOLOv9-t [30]	83.1	63.4	2.6	16.9	86.1
YOLOv10-n [16]	87.8	69.2	2.6	12.2	92.8
YOLOX-t [31]	82.3	52.6	5.0	11.8	87.9
RTMDet-t [32]	90.8	71.2	4.9	12.5	108.5
Faster R-CNN [25]	90.1	67.5	41.3	134.0	60.4
AF-OSD [12]	89.7	-	192.0	136.0	-
R3Det [34]	89.3	-	227	-	-
GRS-DET [35]	89.6	-	200	-	-
OptiSAR-Net (ours)	**91.3**	**75.0**	2.7	11.8	89.8

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TABLE VII
COMPARISON WITH OTHER MODELS ON HRSID

Model	AP_{50}^{val} (%)	$AP_{50:95}^{val}$ (%)	#Param. (M)	FLOPs (G)	$FPS_{b=1}^{3090}$ (f/s)
YOLOv8-n (base)	89.5	64.9	3.0	12.8	**110.5**
YOLOv5-n	88.6	62.9	**2.5**	11.2	107.1
YOLOv9-t [30]	85.1	59.4	2.6	16.9	88.6
YOLOv10-n [16]	88.9	64.0	2.6	12.4	91.7
PPYOLOE-Plus-s [33]	89.6	65.2	7.9	13.6	92.3
RTMDet-t [32]	86.9	61.6	4.9	12.5	102.8
Faster R-CNN [25]	74.7	55.2	41.3	134.0	55.7
SA-YOLOv5-Plus [11]	**94.8**	-	*53.1	*210.4	*47.6
YOLO-SARSI [36]	89.3	64.0	18.43	-	-
CSEF-Net [37]	90.6	-	37.3	104.1	-
DBW-YOLO [38]	88.8	57.4	*6.2	***10.8**	*81.9
OptiSAR-Net (ours)	90.2	**65.4**	2.7	11.8	90.6

TABLE VIII
COMPARISON WITH OTHER MODELS ON SSDD

Model	AP_{50}^{val} (%)	$AP_{50:95}^{val}$ (%)	#Param. (M)	FLOPs (G)	$FPS_{b=1}^{3090}$ (f/s)
YOLOv8-n (base)	97.5	70.4	3.0	8.1	120.1
YOLOv5-n	97.3	70.2	2.5	**7.1**	**116.3**
YOLOv9-t [30]	95.8	68.9	2.6	10.7	94.8
YOLOv10-n [16]	96.8	66.8	2.6	7.8	96.6
YOLOX-t [31]	93.6	54.6	5.0	7.6	93.5
RTMDet-t [32]	96.5	64.2	4.9	8.0	117.5
Faster R-CNN [25]	93.3	63.1	41.3	90.9	64.8
SA-YOLOv5-Plus [11]	98.8	-	*53.1	*134.5	*54.2
DBW-YOLO [38]	**99.2**	-	*6.2	*6.9	*85.8
OptiSAR-Net (ours)	97.6	**70.7**	**2.7**	7.5	95.7

TABLE IX
COMPARISON WITH OTHER MODELS ON CDHD

Model	OPTICAL (HRSC2016)		SAR (*HRSID)		ALL		#Param. (M)	FLOPs (G)	$FPS_{b=1}^{3090}$ (f/s)
	AP_{50}^{val}	$AP_{50:95}^{val}$	AP_{50}^{val}	$AP_{50:95}^{val}$	AP_{50}^{val}	$AP_{50:95}^{val}$			
YOLOv8-n (base)	90.8	74.2	78.7	51.5	84.8	62.9	3.0	12.6	**109.5**
YOLOv5-n	89.4	70.6	79.9	52.1	84.7	61.4	**2.5**	**11.2**	104.7
YOLOv9-t [30]	83.4	64.3	73.0	48.6	78.2	56.5	2.6	16.9	86.5
YOLOv10-n [16]	86.4	67.8	71.6	47.5	79.0	57.6	2.6	12.2	90.8
YOLOX-t [31]	83.2	55.6	67.5	41.9	75.4	48.8	5.0	11.8	88.1
RTMDet-t [32]	91.3	74.6	72.9	48.2	82.1	61.4	4.9	12.5	105.9
PPYOLOE-Plus-s [33]	91.6	74.8	78.2	52.2	84.9	63.5	7.9	13.6	91.2
Faster R-CNN [25]	90.9	69.3	67.4	46.6	78.9	58.0	41.3	134.0	56.2
OptiSAR-Net (ours)	**91.8**	**76.7**	**82.1**	**54.7**	**87.0**	**65.7**	2.7	11.8	89.8

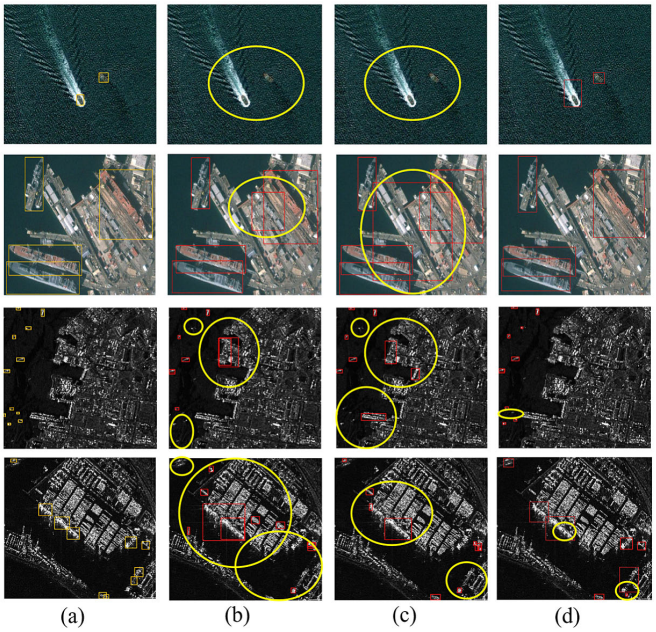


Fig. 10. Visualization of the detection results on the CDHD dataset via various methods. (a) Ground truth. (b) YOLOv8-n. (c) YOLOv10-n. (d) OptiSAR-Net. Yellow circles represent missed or misdirected ship targets.

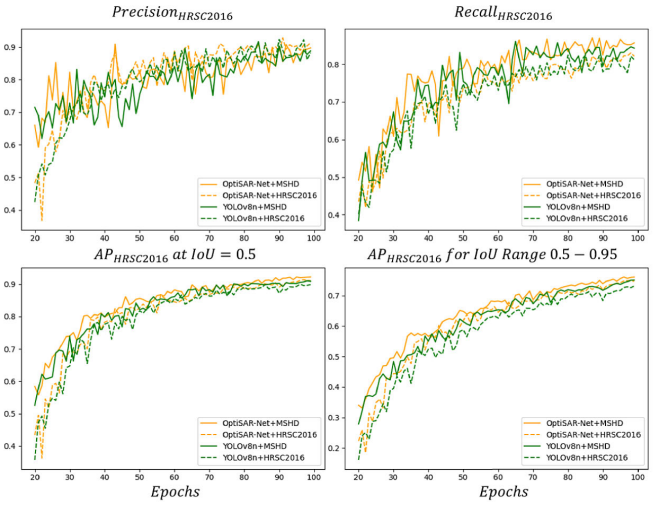


Fig. 11. Performance evaluation of our approach and the baseline model on the same optical images during training on the HRSC2016 and CDHD datasets (first 20 rounds omitted).

mechanisms excel at modeling long-range dependencies, often requiring more raw data for convergence. Despite this constraint, OptiSAR-Net outperforms numerous models, with an accuracy of 97.6%. The precision of DBW-YOLO has surpassed 0.99, achieving SOTA performance on the SSDD dataset. Detection errors are common in complex nearshore scenarios, yet our model maintains high accuracy, as depicted in Fig. 9.

The OptiSAR-Net's multisource detection performance was evaluated on the heterogeneous CDHD dataset, and compared against various models, with the results detailed in Table IX. Models trained on the multidomain dataset demonstrated superior performance in single-domain evaluations on the HRSC2016 dataset compared to those trained solely on single-domain data. For instance, the mAP of Faster R-CNN [25] improved from 67.5% to 69.3%. However, this improvement came at the cost of reduced SAR domain feature perception. In contrast, OptiSAR-Net introduced a dynamic attention mechanism to capture the feature differences between optical and SAR images, effectively utilizing their complementary properties. This approach achieved the best multisource classification performance across both the optical and SAR domains. Compared to baseline models, OptiSAR-Net maintains sensitivity to optical images and achieves a 6.2% increase in mAP for SAR ship detection. The visual comparative results in Fig. 10 clearly illustrate that OptiSAR-Net meets the practical requirements of the multisource detection task more effectively than other models.

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To further illustrate the advantages of cross-domain detection, we visualized the training processes of YOLOv8-n and OptiSAR-Net on the heterogeneous dataset CDHD and the single-domain dataset HRSC2016 in Fig. 11. The evaluations presented are all based on HRSC2016. The models trained on the heterogeneous dataset converge much faster than those trained solely on single-domain data. The increased number of data samples provides the models with more opportunities to learn ship features, even if additional SAR image samples are included, ultimately leading to higher model accuracy. Moreover, OptiSAR-Net, which incorporates attention mechanisms, exhibits more robust cross-domain perception capabilities.

V. Conclusion

This article introduces an end-to-end cross-domain multisource ship detection network called OptiSAR-Net. Drawing inspiration from the human higher visual cortex's attention and goal-oriented recognition mechanisms, we guide the model to focus on ship-related visual features across different frequency domain images, extracting consistent shape, texture, or contrast information. Specifically, the DAA extracts standard features of cross-domain images through secondary attention, providing the backbone with dynamic adaptive feature extraction capabilities. In the SPPF, a novel sparse attention mechanism, BRDA, is introduced to achieve deformable spatial pyramid pooling. It allows the model to flexibly adapt to multiscale variations in ships and reduce its focus on background features from different domains. The VSSA facilitates a lightweight design in the model's neck section while effectively filtering out SAR noise interference. Results indicate that cross-domain multisource detection offers broader application scenarios and potential performance enhancements compared to single-domain detection. To our knowledge, we are the first to explore cross-domain multisource object detection.

This study focuses on utilizing the robust feature extraction capabilities of CNNs and associative learning through attention mechanisms, enabling the model to interpret cross-domain images similar to human understanding. However, the imbalanced multisource data samples, often constrained by practical limitations, have yet to be fully addressed. Looking ahead, more efficient attention mechanisms could integrate multiple data sources, including optical, SAR, and infrared, into cross-domain multisource ship detection tasks. There is even the potential to eliminate the need for distinguishing between different perspectives (e.g., camera views) or target types.

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